



Design of self-heating battery power supply for Antarctic scientific instruments

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HIGHLIGHTS

- Designed a self-heating battery unit for Antarctic scientific instruments.
- Developed an electro-thermal coupling model for fast long-term battery simulation.
- Optimizing heating strategy for maximum capacity gain Using IGA.
- The proposed battery unit can extend discharge duration more than 90 % at $-40\text{ }^{\circ}\text{C}$.

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ABSTRACT

In Antarctic scientific research, low temperatures severely degrade the performance of battery-powered observation instruments, reducing discharge capacity. This paper proposes a self-heating battery module designed for Antarctic applications. A sandwich-style heating structure is designed and the corresponding electro-thermal coupling model is established. A heat exchange mechanism is adopted to efficiently couple electrical and thermal characteristics and reduce computational complexity. A heating strategy that combines the battery's self-heating during discharge with external heating from a film is developed. Integer Genetic Algorithm (IGA) is adopted to optimize the design and strategy to maximize the battery's incremental capacity gain. Validation tests are carried out on a scaled-battery system. By examining its electrical and thermal properties, the heating performance of the batteries under different temperatures and discharge modes is studied. The proposed design significantly extends battery discharge duration in Antarctic environments, achieving an 114 min (>90 %) increase in discharge capacity at $-40\text{ }^{\circ}\text{C}$ compared to unheated setups. This research presents a practical solution for battery thermal management in ultra-cold conditions.

Nomenclature

U_{ocv}	open circuit voltage of battery	λ_i	thermal conductivity of the insulating material
R_0	ohm internal resistance	S_i	surface area of the battery's side
R_1	polarized internal resistance	Q_{bat}	battery self-generated heat
C_1	polarized capacitance	Q_L	heat generated by heating film
T	battery temperature	Q_{dis}	heat dissipation
OCV	Open circuit voltage	Q_J	Joule heat
SOC	state of charge	Q_P	polarization reaction heat
U_0	voltage of internal resistance R_0	Q_R	reaction heat

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U_1	voltage of polarized internal resistance R_1	h	surface heat transfer coefficient of the battery
I, U	battery voltage and discharge current	A	heat exchange area of the battery
R_b	internal thermal resistance of battery	m	mass of the battery
C_b	thermal capacitance of battery	C_{bat}	heat capacity of the battery
P_b	heat generation power of the battery	C_{gain}	incremental capacity gain
P_h	heat generation power of heating film	C_{Th}	total capacity of the heated battery

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(continued)

P_c	power of the monitor and controller	C_{Tu}	total capacity of the unheated battery
T	temperature	SOC_{t0}	state of charge at the initial time (t_0) of battery
T_{bi}	temperature of the i th battery cell	Q_E	summation of the power consumed by the heating film, monitor and controller
T_h	heating temperature of the heating film	Q_T	the power consumption of the whole heating strategy
T_{out}	ambient temperature	U_h, I_h	voltage and discharge current of the heated battery
R_{outj}	thermal resistances between the environment and the j th battery	U_u, I_u	voltage and discharge current of the unheated battery
L_b	battery thickness	C-rate	the charge or discharge current relative to the battery's nominal capacity
λ_b	the battery's thermal conductivity	PI	Polyimide (heating film material)
S_b	the maximum surface area of the battery	HPPC	Hybrid Pulse Power Characterization (a battery testing method)
L_i	thickness of the insulating box	IGA	Integer Genetic Algorithm

1. Introduction

Antarctica [1,2], located at the southernmost tip of the Earth, is recognized as the coldest continent on the planet, with the majority of its surface covered by extensive ice sheets. Within this icy realm, the Ross Sea region [3] represents a pivotal area of study where the lithosphere, cryosphere, biosphere, and atmosphere intersect, creating a complex and dynamic natural system. This region offers a unique opportunity for scientific research and many scientific observation instruments have been installed [4,5]. The energy requirements for these scientific instruments are relatively low and they are supplied with batteries. However, their working environment temperatures can reach as low as -40°C [6], posing a significant challenge to the battery. Such extreme cold notably decreases batteries' charge and discharge efficiency, useable capacity, and power output [7–9]. In order to enhance the performance of the battery in low-temperature environments, research efforts are primarily focused on improving the structural design and materials of batteries [8,10]. From the view of application, the development of effective battery thermal management strategies [11–13] is more practical.

The heating of the battery is divided into external heating and internal heating. The external heating method usually uses heating sheets or heating films to heat the battery, and typically relies on an external power source to drive the heating device [14,15]. Huang et al. [16] introduced the independent thermal management of the battery pack by driving the heating sheet with the battery's own electric energy. Guo et al. [17] used the waste heat recovery of the driving system in the new energy vehicle for battery and cabin heating. Qu et al. [18] proposed the pulse self-heating strategy which is able to quick and safe warming of lithium-ion battery at low temperature and alleviate the battery degradation during heating. External heating is simple and direct, and is generally suitable for batteries with small volume and flat shape. Lin et al. [19] installed the heating film between batteries so that the sides of each battery can be heated and reduce the temperature gradient inside the battery. Jiang et al. [20] designed an internal self-heating strategy to reduce lithium-ion fading through superimposing the discharge current on alternating current for self-heating. Ruan et al. [21] optimized the direct-current discharge heating using a multi-objective genetic algorithm to reduce battery capacity degradation. Guo et al. [22] adopted the optimization on the coupled electro-thermal model to find the optimal preheating strategy. While, these studies use energy from external power sources to heat the battery, which is not suitable for

battery systems without external power sources. Moreover, external heating can easily lead to thermal runaway of the battery [23].

The internal heating method controls the battery discharge method, and uses the internal impedance of the battery to generate heat during the discharge process [24,25]. Qin et al. [26] applied alternating current to the battery, and quickly heated it through a periodic charge and discharge process. Xiong et al. [27] used an external resistance and MOSFET circuit to generate a pulse current to preheat the battery, and based on multi-objective optimization, adjusted the current amplitude and duty cycle of the battery pulse discharge heating to balance the heating rate and the battery aging caused by the heating process. Ruan et al. [28] optimized the control of the external heating resistance, and balanced the four heating objectives of heating time, heating efficiency, battery aging degree and heating uniformity to obtain the optimal heating scheme. Compared with external heating, the heat transfer of internal heating is more uniform, but it involves more considerations, and the design of the discharge circuit and the corresponding control algorithm are more complex [29].

Some studies have concentrated their efforts on refining the intrinsic architecture of the battery. Wang et al. [11] added the direct heating of the electrolyte and electrode inside the battery by burying heating metal boards within the battery. Zhang et al. [30] buried copper tubes with internal heating bars into the battery, which effectively improved the heat transfer and heating effect of the battery, and improves the low temperature performance and heat dissipation capacity of the battery. Hailemichael and Ayalew [31] combined exogenous heating with internal pulse current, combining the advantages of the two, to obtain a more uniform heating performance. However, these methods have a high economic cost and are still in the experimental phase. Meanwhile, these studies aim to enhance the preheating [16] before the battery starts, but do not consider the extent of battery discharge capacity, therefore, the battery capacity will reduce due to the cost of the preheating.

Different from previous works, the design target of the battery heating for Antarctic observation instruments is to increase discharge capacity under low temperature. Therefore, this paper designs a battery heating structure as well as a composite heating strategy for Antarctic applications. A sandwich-style heating structure is designed, where a heating film is sandwiched between every two batteries. It fully utilizes the self-heating during battery discharge, taking the joule heat generated during the battery's own discharge process as internal self-heating. To obtain the optimal design parameters, an efficient electro-thermal coupling method is proposed that can calculate the battery life at different temperatures rapidly and accurately. With the assistance of the Integer Genetic Algorithm and the model, optimum design parameters are automatically obtained. The proposed system achieves an effective enhancement of battery endurance with minimal control.

2. Battery design for Antarctic observation instruments

Scientific observation instruments in Antarctica are extensively utilized for the continuous monitoring of environmental parameters. These instruments typically include a main unit, an array of environmental sensors, and a satellite communication module. The power supply for these systems predominantly relies on batteries, which face significant challenges due to the extremely low-temperature conditions, thereby affecting the system's operational endurance.

In the Antarctic context, conventional power supply solutions do not incorporate heating mechanisms or temperature regulation for batteries. To enhance the discharge capacity of the batteries, this study introduces an innovative battery configuration as illustrated in Fig. 1. The proposed system incorporates a series arrangement of multiple battery units, with each unit featuring a pair of batteries separated by a polyimide (PI) heating film. The batteries are flat and relatively thin, ensuring the heat from the heating film distributes evenly within the batteries, and most of the heat being absorbed by the batteries. This heating film serves a

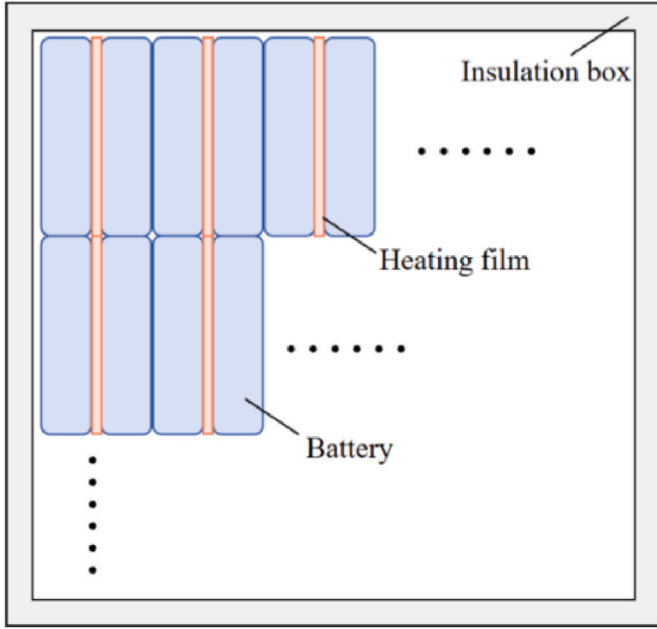


Fig. 1. Schematic of the self-heating battery power supply system. The system comprises parallel-connected battery units. Each battery unit consists two batteries and a heating film sandwiched between them to enable a uniform heating.

critical function by elevating the temperature of the batteries, which is imperative for prolonging their discharge duration. To further optimize energy conservation, the entire battery assembly is encased in insulating materials to reduce thermal loss.

Building on this design, the present paper advances targeted strategies for battery state estimation and thermal management, tailored to the requirements of Antarctic scientific observations. These strategies are designed to maximize the efficiency and reliability of the power supply system under the challenging environmental conditions prevailing in Antarctica.

3. Electro-thermal modeling of the system

To accurately simulate battery performance in the Antarctic environment, an accurate electro-thermal model is essential. There are two main types of electro-thermal models [32]. Physics-based models [33, 34] solve fundamental partial differential equations that describe the electrochemical processes within the battery. While these models offer in-depth understanding of underlying physical processes and degradation mechanisms, they are time-consuming and not suitable for long-term simulations. On the other hand, equivalent circuit models [35–40] represent the battery as an electrical-thermal coupled circuit. These models are computationally fast, making them well-suited for long-term evaluations. In this work, an electro-thermal equivalent circuit model is applied.

The overall flow chart of the model is shown in Fig. 2. The electro-thermal modeling framework begins with independent experimental extraction of the battery's thermal and electrical parameters. The structural thermal model and electric circuit are then integrated into a coupled simulation framework. To ensure computational efficiency, an iterative sequential coupling approach is employed. At each time step, the electric circuit first computes electrical variables and the resulting heat generation. This heat generation is input to the thermal model to update the spatial temperature distribution. The updated temperature is subsequently fed back into the electric circuit model for the next iteration, establishing a bidirectional feedback loop between electrical performance and thermal state. This method balances accuracy and

computational tractability, enabling simulations of extended operational durations while capturing critical electro-thermal interdependencies.

3.1. Battery equivalent circuit

Lithium-ion batteries' performance is sensitive to ambient temperature. When the temperature decreases, the internal resistance of the battery gradually increases, which causes the battery heat generated by the Joule thermal effect to increase [41]. Considering the computation efficiency and accurate modeling of the discharge and heat generation characteristics of batteries, while minimizing computational costs, the first-order RC equivalent circuit [42] is adopted to model the battery.

The parameters required for the construction include the battery opening voltage U_{ocv} , the ohm internal resistance R_0 , the polarized internal resistance R_1 and the polarized capacitance C_1 . These values will change with the change of battery temperature (T) and state of charge (SOC). Therefore, in order to obtain the corresponding parameter values at different temperatures and SOC states, a table for each parameter is established.

It is assumed that the battery discharge current is i , and the equivalent circuit of the battery is

$$U_0 = iR_0(SOC, T) \quad (1)$$

$$U_1 = iR_1(SOC, T) \left[1 - e^{\frac{-t}{R_1(SOC, T)C_1(SOC, T)}} \right] \quad (2)$$

where U_0 and U_1 are voltages of internal resistance R_0 and polarized internal resistance R_1 respectively, and the total voltage of the battery U is

$$U = U_{ocv}(SOC, T) + U_0 + U_1 \quad (3)$$

According to formulas (2) and (3), the voltage of different T and SOC is obtained through battery discharge experiments and parameter extraction. During the simulation, the model adjusts the U_{ocv} and R_0 values based on the battery temperature T and SOC dynamically, so as to achieve the calculation of the battery's thermal coupling.

3.2. Thermal modeling of insulation box

In cold temperature battery thermal management systems, the synergy of insulation materials and heating systems plays a vital role in maintaining battery temperature [43]. To simulate the thermal effects during power discharge, it is necessary to perform thermodynamic modeling of the battery and insulating materials. To simplify the calculation, we propose a thermal equivalent circuit model for the system.

The battery body is wrapped in insulation material with uniform thickness and density, and the external environment temperature is considered constant. The batteries in the battery box are paired with each other, and a PI heating film sandwiched between the two batteries in each group. Assuming uniform heat generation and propagation within the battery [44], the thermal equivalent circuit model for an individual battery unit is shown in Fig. 3. In this model, R_b , C_b , and P_b represent the internal thermal resistance, thermal capacitance, and heat generation power of the battery, respectively. T_{bi} (where $1 \leq i \leq 2n$) denotes the internal temperature of the i th battery cell from left to right. P_h and T_h correspond to the heat generation power and heating temperature of the heating film. R_{outj} and $R_{out(j+1)}$ are the thermal resistances between the outer environment and the left and right sides of the j th battery group, respectively (where $1 < j < n+1$). Finally, T_{out} denotes the ambient temperature. Considering heat propagation only along the thickness direction of the battery, the relationship between the battery thermal resistance R_b and the battery's thermal conductivity is given by:

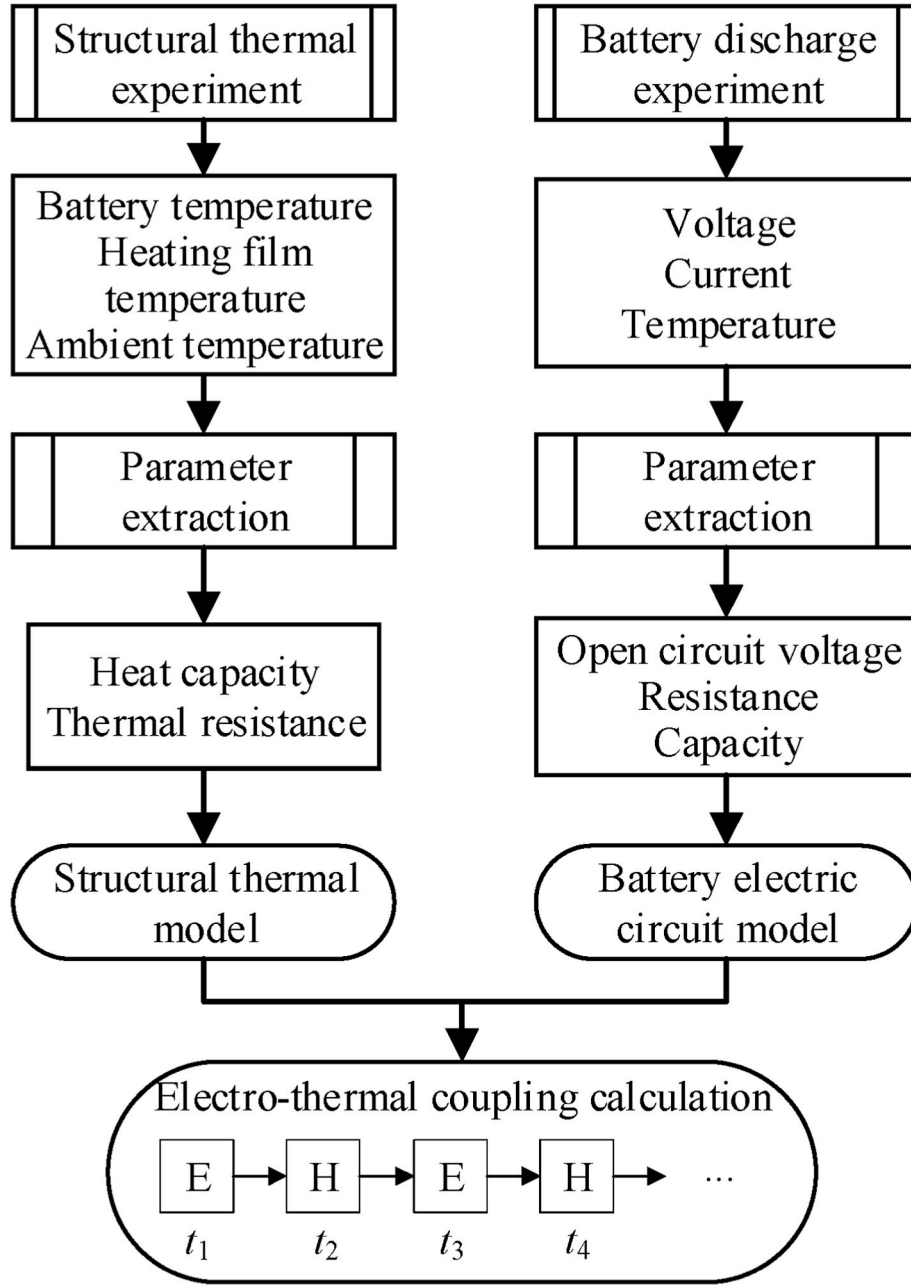


Fig. 2. Electro-thermal coupling system modeling and experiment flowchart.

$$R_b = \frac{L_b}{2\lambda_b S_b} \quad (4)$$

where L_b represents the battery thickness, λ_b corresponds to the battery's thermal conductivity, and S_b denotes the surface area of the battery. Similarly, we have

$$R_{out1} = R_{out(n+1)} = \frac{L_i}{\lambda_i(2S_b + S_i)} \quad (5)$$

$$R_{outj} = \frac{L_i}{\lambda_i S_i} \quad (6)$$

in which L_i represents the thickness of the insulating box, λ_i corresponds to the thermal conductivity of the insulating material, and S_i denotes the surface area of the battery's side, specifically the area in contact with the insulating box.

The relationship between the battery, heating film, and insulating

layer allows us to express the heat transfer for the first battery cell as follows:

$$C_b \frac{dT_{b1}}{dt} = P_b + \frac{T_{out} - T_{b1}}{R_{out1} + R_b} + \frac{T_h - T_{b1}}{R_b} \quad (7)$$

For the 2nd to the $(n-1)$ th batteries, which are in contact with batteries on both sides, their thermal transmission relationships are expressed as:

$$C_b \frac{dT_{b(2j-1)}}{dt} = P_b + \frac{T_{b(2j-2)} - T_{b(2j-1)}}{2R_b} + \frac{T_{out} - T_{b(2j-1)}}{R_{out} + R_b} + \frac{T_h - T_{b(2j-1)}}{R_b} \quad (8)$$

$$C_b \frac{dT_{b(2j)}}{dt} = P_b + \frac{T_{b(2j+1)} - T_{b(2j)}}{2R_b} + \frac{T_{out} - T_{b(2j)}}{R_{out} + R_b} + \frac{T_h - T_{b(2j)}}{R_b} \quad (9)$$

$$\frac{T_j - T_{out}}{R_{out2}} = \frac{T_{b(2j-2)} - T_j}{R_b} + \frac{T_{b(2j-1)} - T_j}{R_b} \quad (10)$$

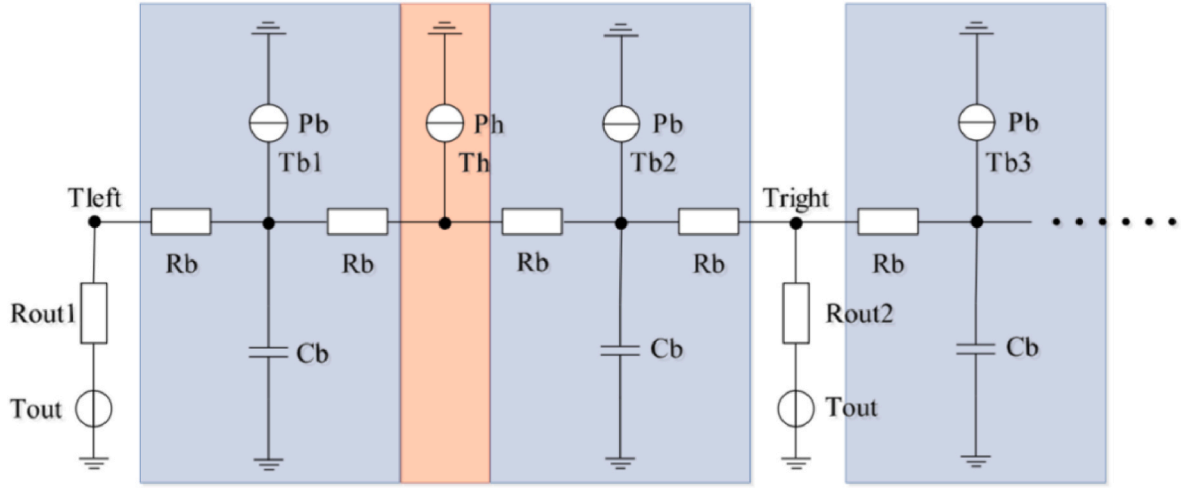


Fig. 3. The equivalent heat circuit for the first battery cell.

From an energy conversion perspective, the heating film consumes and converts it into heat energy for both the heating film itself and the heated object. Considering that the heating film operates based on a fixed temperature threshold control, we can assume that the heating film power varies with the battery's temperature. Therefore, the output power of the heating film can be equivalently represented as:

$$P_h = \frac{T_h - T_{b(2j-1)}}{R_b} + \frac{T_h - T_{b(2j)}}{R_b} \quad (11)$$

3.3. Electro-thermal coupling of the system

As shown in Fig. 4 is the electric-thermal coupled system, the heat generation components consist of the self-generated heat (Q_{bat}) from within the battery during discharge and the external heating film heat (Q_L)

$$Q_{bat} = P_b \quad (12)$$

$$Q_L \approx P_h \quad (13)$$

The heat dissipation part is the heat exchange between the battery and the insulation material, and the insulating material and the external environment, with the total amount represented by Q_{dis} . The self-heating inside the battery is divided into reaction heat, side reaction heat, polarization reaction heat, and Joule heat. Among them, the reaction heat and side reaction heat are the entropy heat caused by the internal chemical reaction during the charging and discharging process of the battery [40]. When the battery is charged, it is manifested as

endothermic, and when discharged, it is manifested as exothermic, also known as reversible heat; while the polarization reaction heat and Joule heat are also called irreversible heat, which is the heat generated by the Joule effect when the current passes through the internal resistance of the battery, which is irreversible. Therefore, the internal heat generation rate of the battery during discharge can be expressed as:

$$Q_{bat} = Q_J + Q_P + Q_R = I^2 R_0 - IT \frac{dU_{OCV}}{dT} \quad (14)$$

where Q_J is Joule heat, Q_P is polarization reaction heat, and Q_R is reaction heat. Since the energy of the reaction heat is small, it can be ignored, then equation (14) simplifies to:

$$Q_{bat} \approx Q_J + Q_P = I^2 R_0 \quad (15)$$

When the battery is working and the temperature rises, it exchanges heat with the surrounding environment mainly in the form of heat conduction and heat convection. When there is no insulation material, the heat exchange is specifically manifested as the heat convection dissipation between the battery and the air; and when the battery is wrapped with insulation material on the outside, it is manifested as the heat conduction dissipation between the battery and the insulation material. Therefore, the heat dissipation rate of the battery Q_{dis} can be expressed as:

$$Q_{dis} = hA(T_b - T_{out}) \quad (16)$$

or

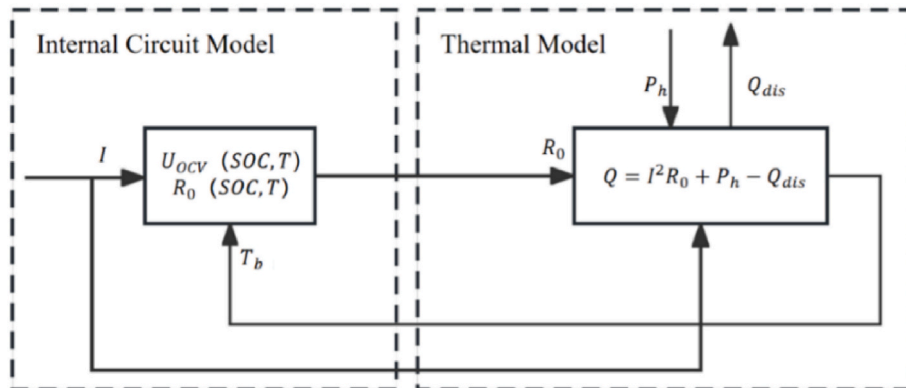


Fig. 4. The mechanism of electro-thermal coupling.

$$Q_{dis} = \frac{\lambda_i}{L_i} A (T_b - T_{out}) = \frac{(T_b - T_{out})}{R_{out} S_i} \quad (17)$$

where h is the surface heat transfer coefficient of the battery, A is the heat exchange area of the battery.

In adherence to the principle of energy conservation, the thermal energy produced by a battery during its discharge process is partitioned into two primary pathways. A portion of this heat is expelled into the surrounding environment, while the remainder is absorbed internally by the battery itself. Considering the situation of external heating film, most of the power consumed by the heating film is converted into heat and absorbed by the battery. Consequently, the thermal behavior of the battery under such conditions can be accurately described by the following thermal model equation:

$$C_{bat} m \frac{dT}{dt} + \frac{(T_b - T_0)}{R_{out} S_i} \approx I^2 R_0 + P_h \quad (18)$$

where m is the mass of the battery, C_{bat} is the specific heat capacity of the battery.

Assuming that the temperature of the heating film is constant, i.e., $dT_h/dt = 0$, and the temperatures of the two batteries in a battery unit are equal, then according to the thermal equivalent circuit model, we have:

$$P_h = \frac{T_h - T_b}{R_b} \quad (19)$$

The principal diagram of the electro-thermal coupling model is shown in Fig. 4, highlighting the interdependence between the electrical and thermal models. Within this framework, key parameters from the electrical model, including the battery's internal resistance and discharge current, are conveyed to the thermal model, influencing the self-heating component (Q_{bat}) therein. Conversely, the thermal model relays the battery's temperature back to the electrical model, integrating with SOC parameters to refine variables such as resistance and the residual capacity available in the electrical model. This proposed thermo-electrical model features a simplified thermo-electrical energy exchange mechanism prioritizing energy balance during long-term discharge, which allows for efficient simulation of the battery's long-term behavior. By inputting intrinsic battery properties and external environmental conditions, the model can efficiently predict key outputs including electrical performance, thermal behavior, and discharge capacity.

4. Self-heating strategy and optimization

In order to increase the reliability of the system and reduce the power consumption of the control system, the battery heating strategy should adopt the simplest possible solution. In order to explore the optimal battery thermal management scheme, this paper uses optimization algorithms to optimize the design of parameters.

4.1. Objective function and constraints

The primary objective of this study is to enhance the operational longevity of the device through the implementation of a self-heating mechanism and insulation for the batteries. Specifically, the aim is to achieve an increase in the battery's capacity post-heating relative to its pre-heating state. To quantify this enhancement, the optimization target is defined as the incremental capacity gain (C_{gain}) of the battery after the heating process. Given that the heating membrane is energized by the battery's own power, it is essential to consider the power consumption dynamics. The optimization goal, therefore, encompasses the net power consumption, which is the difference between the available power after the battery's performance and the power consumed by the internal heating mechanism. This approach ensures an efficient balance between

the thermal management of the battery and the device's operational efficiency. The expression is:

$$C_{gain} = (C_{Th} - C_{Tu}) SOC_{t_0} - Q_E \quad (20)$$

where C_{Th} is the total discharge capacity of the battery after the battery is heated, and the C_{Tu} is the total discharge capacity of the unheated battery. SOC_{t_0} is the state of charge from the battery starts, the battery discharge starts from full charge, so the initial value of $SOC_{t_0} = 1$. Q_E is the summation of the power consumed by the heating film, monitor and controller. Here discharge capacities can be further written as:

$$C_{Th} \cdot SOC_{t_0} = \int_0^{t_{end2}} U_h I_h \quad (21)$$

$$C_{Tu} \cdot SOC_{t_0} = \int_0^{t_{end1}} U_u I_u \quad (22)$$

where t_{end2} is the end of the battery pack discharge. In addition, if the battery group does not perform heating throughout the process, the total discharge time is t_{end1} . UI is the product of the battery voltage and discharge current, that is, the output power. $U_h I_h$ And $U_u I_u$ are the output power of the heated and unheated battery respectively.

As shown in the flowchart in Fig. 5, the strategy is designed following the rules to start heating at a specific time to maintain the temperature of the heating film, continue heating the battery until it is depleted, or the output power drops below the minimum operating power of the load. Assume that the power of the monitor and controller is P_c , the consumption of heating strategy is

$$Q_E = \int_{t_{start}}^{t_{end2}} P_h + \int_0^{t_{end2}} P_c \quad (23)$$

where t_{start} is the time when the external heating starts. Based on the previous analysis, assuming a uniform battery temperature (T_b) on both sides of the heating membrane, substituting equation (19) into equation (23), we obtain:

$$Q_E = \int_{t_{start}}^{t_{end2}} 2 \frac{T_h - T_b}{R_b} + \int_0^{t_{end2}} P_c \quad (24)$$

Consequently, the target function of the heating strategy (equation (20)) can be rewritten using equations (21), (22) and (24) as:

$$\max C_{gain} = \int_0^{t_{end2}} U_h I_h - \int_0^{t_{end1}} U_u I_u - \int_{t_{start}}^{t_{end2}} 2 \frac{T_h - T_b}{R_b} - \int_0^{t_{end2}} P_c \quad (25)$$

According to the heating strategy, t_{start} and T_h are decision variables, and the lower limit of T_h is set to $\max_{0 < t < t_{end1}} T_b$, that is, the heating power P_h cannot be less than 0, and the battery temperature after heating cannot be lower than the battery temperature before heating. In addition, considering that there is an upper limit on the heating power of the heating film, and there are safety risks if the heating temperature is too high, the heating film temperature cannot be too high. Depending on the ambient temperature and discharge conditions, the upper limit of T_h also changes dynamically. Therefore, it is limited to:

$$0^\circ C \leq T_h - \max_{0 < t < t_{end1}} T_b = P_h R_b \leq 30^\circ C \quad (26)$$

4.2. Integer genetic algorithm

In order to explore the optimal battery insulation and heating strategy, the integer genetic algorithm (IGA) [45] is used to optimize the parameters. Genetic algorithms (GAs) model the process of natural evolution, embodying the principle of the survival of fittest. These algorithms function within a set of rules for iterative evaluation, predisposing them to favor superior solutions. The screened solutions are subsequently added to a mating pool, whence a new generation of solutions is produced. This continuous process of inter-generational

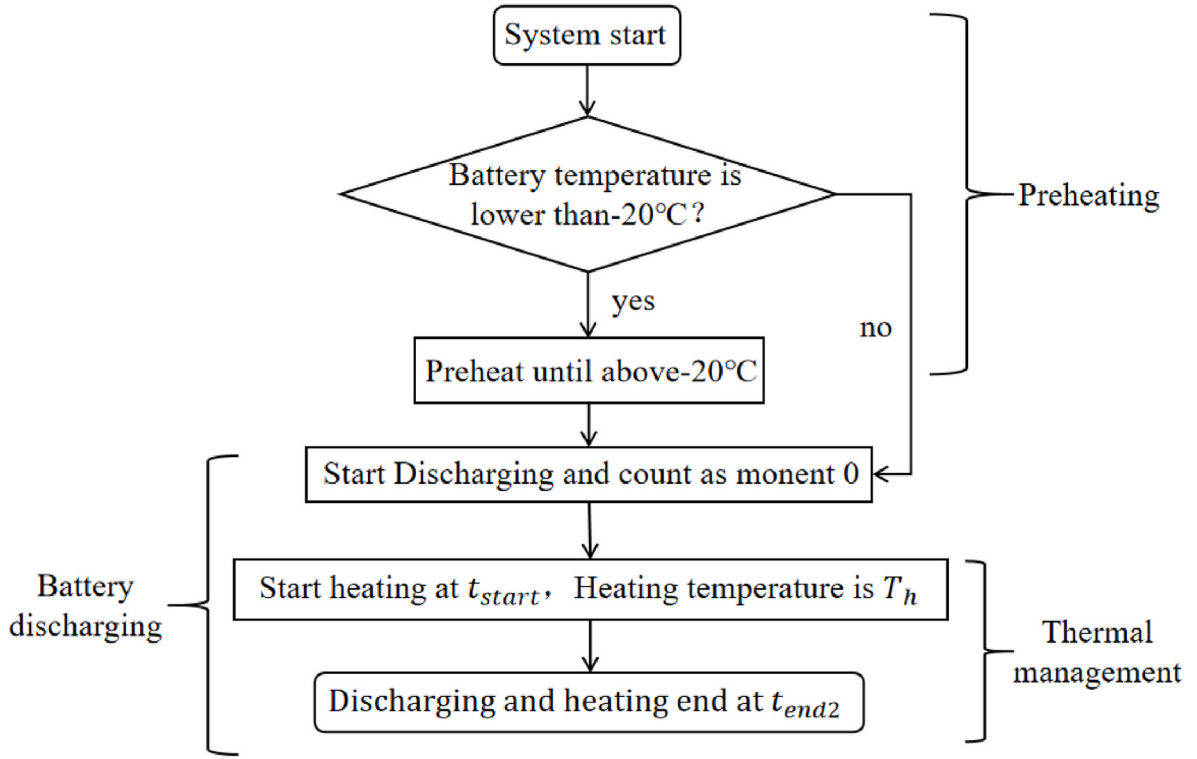


Fig. 5. Battery heating strategy flowchart.

screening enables the selection of the individual demonstrating the highest adaptation, thereby facilitating the extraction of the optimal solution.

Considering manufacturing precision constraints, the variable values for temperature and other pertinent parameters within the heating optimization model are restricted to integer values. Consequently, a truncation process is applied to conform to these limitations as follows:

If x_i is an integer, then $\bar{x}_i = x_i$; otherwise, $\bar{x}_i$ is equal to $[x_i]$ or $[x_i] + 1$, where $[x_i]$ is the maximum integer smaller than x_i , and the probability of the two cases is equal.

This approach adds randomness of the generated next generation of subsets, avoiding the possibility of generating the same integer value when the real values between the same two continuous integers are cut off. Therefore, the optimization process of the IGA in this study is:

- 1) Generate enough initial solutions, whose the values of variables need to meet the specified boundary conditions and integer restrictions;
- 2) Check whether the termination conditions are met, and then terminate, otherwise proceed to step (3);
- 3) Filter the current population, and form the selected results for forming the mating pool;
- 4) Individuals in the mating pool cross and mutate a certain probability to form a population of the next generation of solution;
- 5) If necessary, add an integer limit on decision-making variables and evaluate the adaptation value;
- 6) Add the new group to the old population and return to the steps (2).

5. Experiment and parameter extraction

The measurement of thermodynamic parameters for batteries and insulation materials is complex. In this study, an electric thermal model is constructed for the battery and insulation enclosure system. Through experiments and parameter fitting techniques, the specific heat capacity, thermal conductivity, and other pertinent thermophysical properties of both the battery and the insulating materials are derived indirectly.

5.1. The configuration of experiment

This study selected an 18650 ternary lithium battery power supply and polyurethane foam for insulation. Their main parameters are shown in Tables 1 and 2:

The experimental configuration is presented in Fig. 6. The battery system was placed in a Meling -45°C low-temperature chamber with the precision of $\pm 1^\circ\text{C}$. To ensure the accuracy of temperature measurements, seven PT100 temperature sensors were installed. Two of these sensors were positioned at the top and bottom of the chamber, one of which was on the top surface of the insulation box, while the remaining four were placed within the battery system, specifically on the surfaces of two batteries, the heating film, and in the air space, all at the central. The data from these temperature sensors were recorded at 1 s intervals using a HEEYII HY-R3004 multi-channel recorder. The electrodes of the battery were connected to a XINDANENG BSDS100-10-20 battery discharge tester. Both temperature and discharge data were stored and processed on a computer. To extract battery model parameters, both Hybrid Pulse Power Characterization (HPPC) and constant current discharge tests were conducted. Additionally, cooling and

Table 1

The parameters of 18650 ternary lithium battery.

Parameter/unit	Value
Battery shape	Cylindrical
Nominal Voltage (V)	3.7 V (per cell); 11.1 V (pack-level nominal voltage for 3 cells connected in series)
Nominal capacity (mAh)	3000
Anode Material	Graphite
Cathode Material	Nickel-Cobalt-Manganese
Electrolyte	Lithium hexafluorophosphate (LiPF_6) in ethylene carbonate/dimethyl carbonate solvent
Charging cut-off Voltage (V)	12.6
Discharging cut-off voltage (V)	8.5

Table 2

The parameters of insulation material.

Parameter/unit	Value
Thermal insulation material	Polyurethane
Material density (kg/m^3)	35
Material heat conduction coefficient ($\text{W/K} \times \text{m}$)	0.024

heating tests were performed to fit the thermal parameters of the system.

To extract battery model parameters, both Hybrid Pulse Power Characterization (HPPC) tests and constant-current discharge capacity tests were conducted. Additionally, heating experiments were performed to fit the thermal parameters of the system. According to the requirements of the Antarctic observation instruments, the rated power of the load is 12.2W, and the discharge current is between 0.75 and 1.12A. Thus, the discharge performance was tested at currents of 0.5 A and 1 A (corresponding to C-rates of approximately 0.17C and 0.33C, respectively), where 1C is defined as a discharge current equal to the battery's nominal capacity. Considering the environmental temperature of the Antarctic site, the experiment temperatures were set at -40°C – 20°C for discharge tests and -40°C to -10°C for heating tests. In the heating tests, the target heating film temperature ranged from 0°C to 30°C . In each test, 4 h were allowed to ensure thermal

equilibrium between the battery and the ambient conditions. The detailed test settings are listed in Table 3.

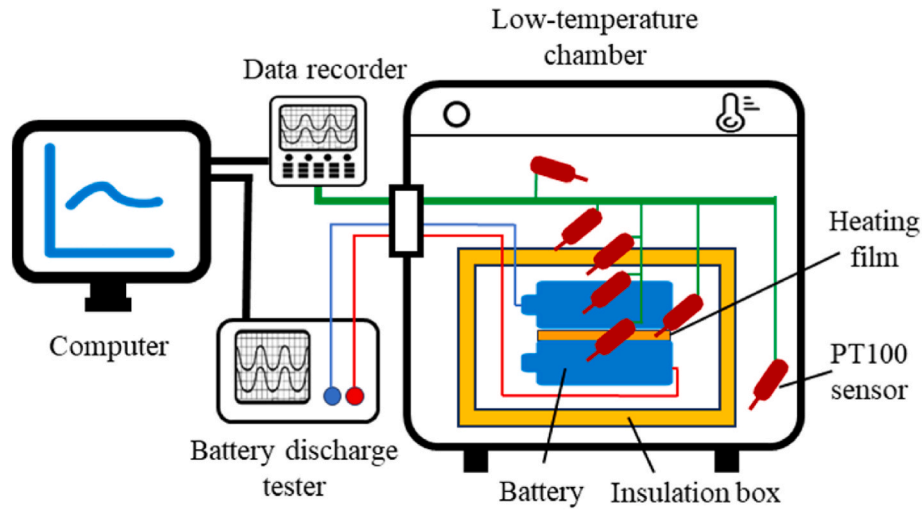
HPPC Test: The HPPC experiment is employed to obtain parameters including the open-circuit voltage, equivalent resistance, and capacitance of batteries. The procedure involves.

- 1) Place the fully charged battery in a low-temperature chamber at different temperatures until the battery temperature stabilizes;
- 2) Discharge the battery at a constant current until 10 % of its SOC is released;
- 3) Pause the discharge and let the battery rest for 1 h to stabilize its voltage;

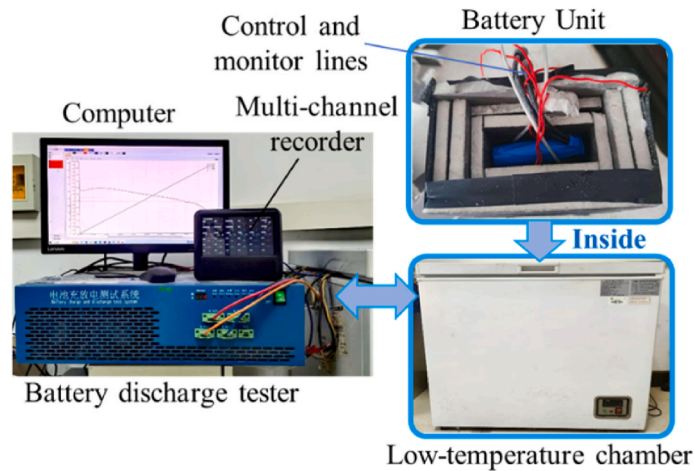
Table 3

The setting of battery discharge experiment.

Experimental type	Environmental temperature/ $^\circ\text{C}$	Discharge current/A	Heating temperature
Constant current discharge	$-40, -30, -20, -10, 0, 10, 20$	0.5, 1	/
HPPC discharge	$-40, -30, -20, -10, 0, 10, 20$	0.5, 1	/
Cooling and heating experiment	$-40, -30, -20, -10$	/	0, 10, 20, 30



(a)



(b)

Fig. 6. The configuration of low-temperature battery discharge experiment. (a) The detailed configuration diagram. (b) Picture of the experiment equipment.

- 4) If the battery can continue discharging, return to Step 2 for another constant current discharge, repeating this cycle until the battery reaches its discharge cutoff;
- 5) Record the discharge current and battery voltage data throughout the process using a charge-discharge tester for subsequent extraction of parameters related to the circuit model.

Constant current discharge test: The battery capacity testing experiment is conducted to determine the maximum available discharge capacity of the battery under specific temperatures and discharge currents. The experiment involves discharging a fully charged battery at various temperatures using different discharge currents until reaching the cutoff voltage. The procedure includes.

- 1) Place the fully charged battery in the low-temperature chamber at different temperatures until the battery temperature stabilizes;
- 2) Discharge the battery at a constant current until reaching the discharge cutoff;
- 3) Record the discharge capacity at the end of the experiment under different temperatures and discharge currents for estimating the remaining available capacity of the battery.

Heating test: In order to fit thermal parameters, it is necessary to conduct heating tests on the system. Multiple combinations of environmental temperatures and heating film temperatures were tested in the experiment. The specific steps are as follows:

- 1) Place the battery wrapped with insulation material in low-temperature chambers at different temperatures to allow it to cool down.
- 2) After the battery temperature stabilizes, turn on the heating film to heat it at a constant temperature until the battery temperature stabilizes again.
- 3) Record the ambient temperature, the temperature at the contact surface between the battery pack and the insulation, and the temperature of the heating film throughout the process. These data are used to extract parameter values such as specific heat capacity and thermal resistance in the thermal model.

5.2. Parameter extraction and validation

In order to extract the electrical and thermal parameters for the battery circuit model [46], the discharge current and ambient temperature of the discharge experiment were used as inputs, and the battery voltage as the reference output response, which were compared with the model voltage output, and the parameters of the battery model were fitted to the U_{OCV} and R_0 of the battery model using nonlinear least-squares optimization. Finally, the above parameter values of the battery under different temperatures and SOC were exported and imported into the battery simulation model in the form of a two-dimensional table as shown in Tables 4 and 5.

The relevant parameters of the extracted thermal model were fitted with the ambient temperature and the heating film temperature as the input, and the battery temperature as the output, and the obtained thermal parameters are shown in Table 6.

Table 4
Open voltage U_{OCV} at different temperatures and SOC (unit: V).

SOC	temperature						
	−40 °C	−30 °C	−20 °C	−10 °C	0 °C	10 °C	20 °C
20 %	9.67	9.77	9.87	9.97	10.07	10.17	10.27
40 %	10.33	10.42	10.51	10.60	10.70	10.80	10.86
60 %	10.81	10.87	10.93	11.02	11.11	11.22	11.29
80 %	11.21	11.33	11.44	11.53	11.63	11.74	11.82
100 %	11.82	12.00	12.17	12.27	12.38	12.49	12.59

Table 5
Internal resistance R_0 at different temperatures and SOC (unit: Ω).

SOC	temperature						
	−40 °C	−30 °C	−20 °C	−10 °C	0 °C	10 °C	20 °C
20 %	7.02	6.16	5.31	4.45	3.59	2.63	1.68
40 %	5.90	4.50	3.10	2.34	1.79	1.59	1.31
60 %	4.04	3.09	2.14	1.73	1.53	1.43	1.29
80 %	3.38	2.75	2.13	1.71	1.51	1.41	1.28
100 %	3.11	2.57	2.03	1.65	1.50	1.41	1.28

In order to verify the accuracy of the simulation model, it is necessary to put the battery into the battery box and then put the battery box into the low-temperature box. After the temperature is stable, the discharge experiment is conducted, and the heating film is opened at some time in the middle, and the heating temperature is random.

The temperature and voltage changes of the battery pack in the battery compartment when it is discharged and heated were verified, and the curves are shown in Fig. 7. It can be seen from the figure that the temperature and voltage curve trends of the simulation and experiment can be basically consistent under different discharge conditions and temperatures, and the absolute deviation is within ± 2 °C and the voltage's absolute deviation is within ± 0.2 V, indicating that the model has sufficient simulation accuracy.

6. Design of self-heating strategy

Heating the battery within an optimal range allows it to release more capacity, leading to maintaining longer working hours. For this design goal, the battery pack should be heated after it starts to discharge and before the discharge ends. The heating temperature is set to a constant, and the optimization is evaluated to find optimal temperature to increase battery discharge capacity. The heating start time (t_{star}) and the temperature of the heating film (T_h) are used as optimization variables, and the net capacity increase (C_{gain}) of the battery after external heating is used as the optimization target. As the battery discharge current increases, the temperature of the battery will also rise during discharge and self-heating, which will weaken the effect of external heating on increasing the battery temperature on battery capacity improvement. It may even occur that the increase in battery capacity cannot make up for the heating energy. Therefore, for the two heating strategies, an integer genetic algorithm is used, combined with certain constraints, for optimization. The IGA optimization in this work was configured with the following settings: a population size of 50, a crossover rate of 0.6, a mutation rate of 0.03, and a maximum generation limit of 200.

Each degree of the heating temperature is taken as 1 unit. Based on the capacity of the heating film, the upper and lower limits of the heating temperature are set to the range of [0, 30] °C. The heating start time is restricted to the interval of [0, 200] minutes. After optimization by IGA, the optimal heating strategy for the battery under different ambient temperatures is obtained and the results are presented in Table 7. Fig. 8 shows the changes in the remaining capacity of the battery during the discharge process before and after the heating strategy is adopted at different ambient temperatures.

By comparing the curves of the battery with (Strategy) and without

Table 6
Thermal parameters of the battery and the insulation box.

parameter	numerical/unit
C_b	$723.77 \frac{J}{kg \cdot K}$
R_b	$1.90 \frac{K}{W}$
R_{out}	$11.48 \frac{K}{W}$

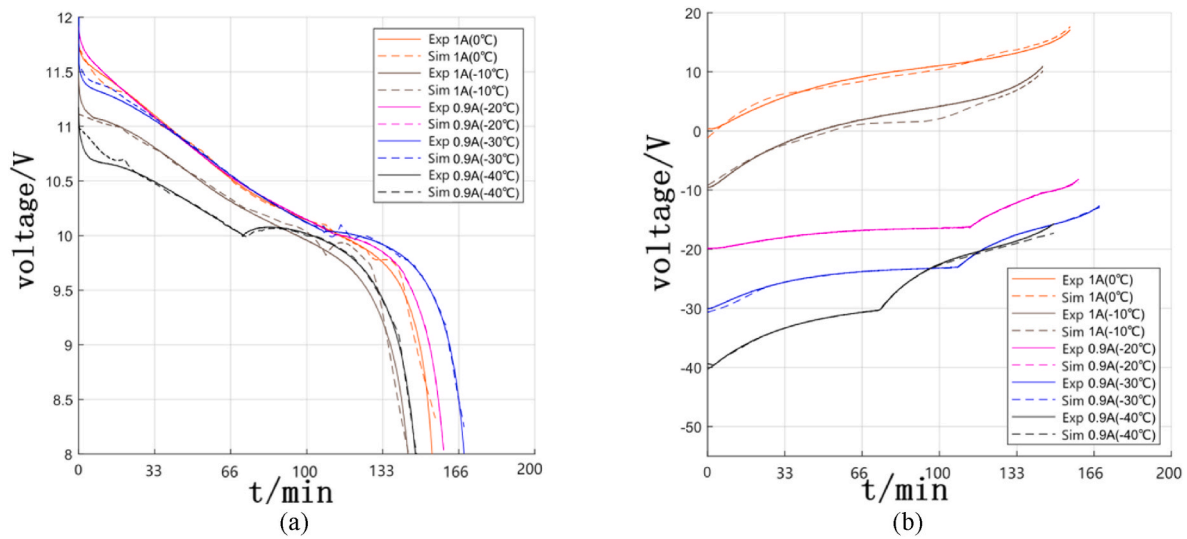


Fig. 7. Comparison of battery low-temperature discharge heating experiment and simulation curve (a) Discharge voltage comparison; (b) Battery temperature comparison.

Table 7

Heating strategy optimization results.

Discharge current/A	Ambient temperature/°C	Heating begins time t_{start}/min	Heating temperature $T_h/^\circ\text{C}$	Battery charge before and after heating C_{T0}/Wh	Heating system consume Q_c/Wh	Net electricity increase ratio (%)
1	-10	96	23	19.62/24.50	2.42	12.5
1	-20	84	20	17.70/23.95	2.50	21.1
1	-30	71	10	16.93/21.79	2.44	21.2
1	-40	60	0	12.80/19.48	2.35	33.8
0.5	-10	200	21	21.08/24.50	4.17	-3.5
0.5	-20	185	20	19.36/24.31	4.32	3.2
0.5	-30	160	10	15.85/22.71	4.25	16.4
0.5	-40	115	0	10.89/21.17	4.25	55.5

(Original) the heating strategy at each temperature, it can be found that at -40°C , -30°C , -20°C , and -10°C , the heating strategy makes the remaining capacity of the battery higher at the same discharge time.

Taking -40°C as an example, during the discharge process, the remaining capacity of the battery with the heating strategy is always higher than that of the unheated battery. This indicates that the heating strategy effectively improves the discharge capacity of the battery at low temperatures, thereby extending the working time of the battery. This is because heating raises the battery temperature, reduces the internal resistance of the battery, and reduces battery polarization, enabling the chemical reactions inside the battery to proceed more smoothly, thus releasing more electricity.

As the ambient temperature increases, the increase in the remaining capacity of the battery brought by the heating strategy varies. At -40°C , the heating strategy has a more obvious effect on increasing the remaining capacity of the battery with a 114 min extension, while at -10°C , the increase is relatively small with only 18 min. This is because the lower the ambient temperature, the more significant the increase in the internal resistance of the battery, and the stronger the inhibitory effect of low temperatures on battery performance. At this time, the heating strategy is more effective in improving battery performance and increasing the discharge capacity. As the ambient temperature rises, the performance of the battery itself improves, and the additional improvement effect brought by the heating strategy weakens accordingly.

7. Conclusion

This paper focuses on developing a power supply for Antarctic

observation instruments, aiming to address the challenge of extending battery discharge duration in cold environments. A thermal coupling model for the battery-heating and insulation-enclosure system was established. Using experimental characterization and parameter fitting, we derived the characteristics of each component and validated the model's accuracy, demonstrating the absolute deviation of temperature prediction is less than 2 %. A self-heating strategy was designed for batteries. It combines internal self-heating of battery during discharge and external heating by a film. Using discharge capacity as the optimization goal and heating temperature and start time as variables, the Integer Genetic Algorithm (IGA) was used to find optimal designs. Experimental validation confirmed the efficacy of the proposed heating strategy: appropriate battery heating significantly extends the discharge duration of power supplies for Antarctic observation equipment. At -10°C , the discharge time increases by 18 min; however, as ambient temperature decreases, the effect of heating on prolonging the battery's useable runtime becomes more pronounced. At -40°C , heating results in a 114 min extension ($>90\%$) in discharge duration.

The work utilizes a simplified model and a reliable, conventional heating setup, potentially limiting dynamic accuracy and excluding advanced heating options. The specific focus on single-discharge endurance also meant that long-term battery degradation effects were not studied in detail. Addressing these limitations, future work includes: 1) Enhance the thermo-electrical model's accuracy (e.g., for dynamic loads, detailed temperature profiles) by incorporating additional physical factors while preserving computational efficiency. 2) Investigate potentially more efficient heating technologies, carefully evaluating the trade-offs between performance gains and the reliability required for demanding applications. 3) Quantify the long-term impact of the

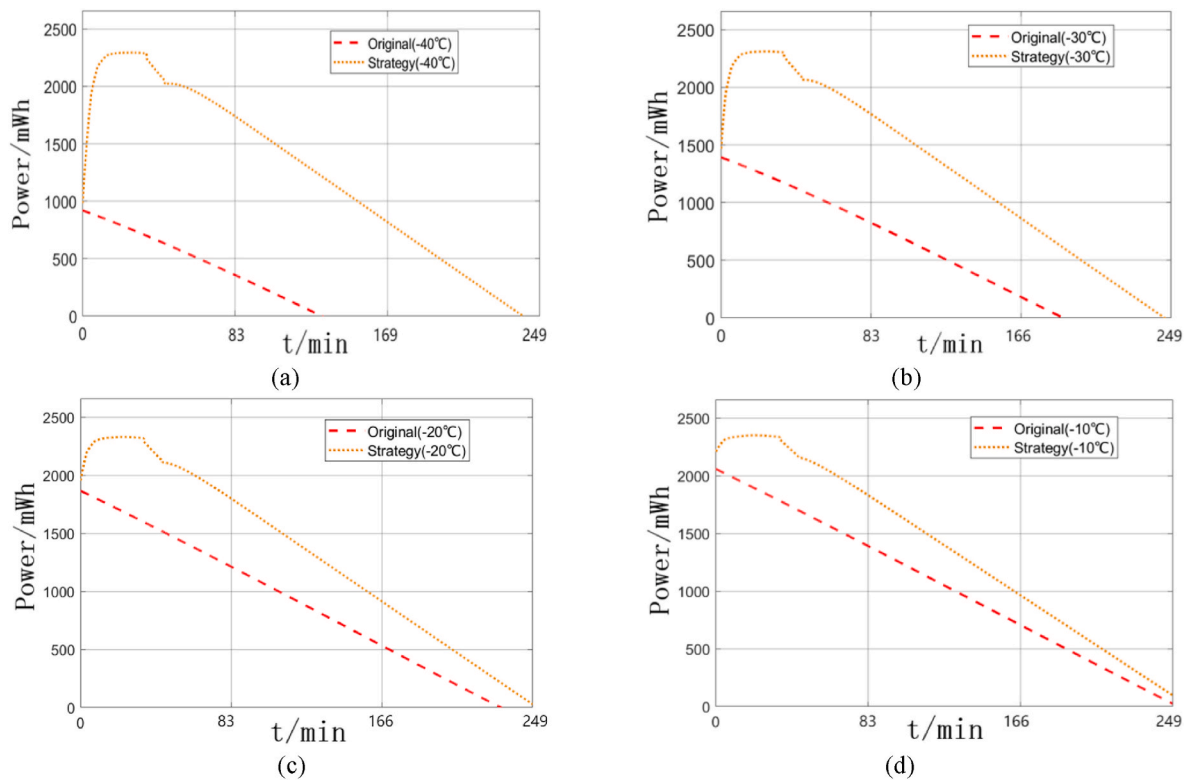


Fig. 8. The residual capacity of the battery discharge process, (a) under -40°C , (b) under -30°C , (c) under -20°C , and (d) under -10°C .

optimized self-heating strategy and prolonged exposure to the Antarctic environment on battery degradation and cycle life through dedicated experimental campaigns and modeling. 4) Develop adaptive heating control strategies that dynamically adjust parameters based on real-time battery state and environmental conditions to potentially achieve further performance optimization compared to pre-set parameters.

CRedit authorship contribution statement

Hong Cai Chen: Writing – review & editing, Methodology, Conceptualization. **Zihan Zhang:** Writing – original draft, Validation, Methodology. **Mingpei Xu:** Writing – review & editing, Visualization, Validation. **Kanjian Zhang:** Supervision, Funding acquisition. **Haikun Wei:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] J. Turner, H. Lu, J. King, et al., Extreme temperatures in the antarctic, *J. Clim.* 34 (7) (2021) 2653–2668, <https://doi.org/10.1175/JCLI-D-20-0538.1>.
- [2] M. Zhang, M. Haward, The Chinese antarctic science programme: origins and development, *Antarct. Sci.* 34 (2) (2022) 191–204.
- [3] R.M. Roura, The XXXVI meeting of the commission for the conservation of Antarctic marine living resources, *Polar J.* 8 (1) (2018) 208–211, <https://doi.org/10.1080/2154896X.2018.1477309>.
- [4] H. Yang, C.A. Kulesa, C.K. Walker, et al., Exceptional terahertz transparency and stability above dome A, Antarctica, *Publ. Astron. Soc. Pac.* 122 (890) (2010/03/19 2010) 490, <https://doi.org/10.1086/652276>.
- [5] W. Saunders, J.S. Lawrence, J.W.V. Storey, et al., Where is the best site on Earth? Domes A, B, C, and F, and ridges A and B, *Publ. Astron. Soc. Pac.* 121 (883) (2009/08/20 2009) 976, <https://doi.org/10.1086/605780>.
- [6] Y.-S. Choi, C.-H. Ho, S.-W. Kim, et al., Observational diagnosis of cloud phase in the winter Antarctic atmosphere for parameterizations in climate models, *Adv. Atmos. Sci.* 27 (6) (2010/11/01 2010) 1233–1245, <https://doi.org/10.1007/s00376-010-9175-3>.
- [7] S.S. Zhang, K. Xu, T.R. Jow, The low temperature performance of Li-ion batteries, *J. Power Sources* 115 (1) (2003/03/27/2003) 137–140, [https://doi.org/10.1016/S0378-7753\(02\)00618-3](https://doi.org/10.1016/S0378-7753(02)00618-3).
- [8] M.-T.F. Rodrigues, G. Babu, H. Gullapalli, et al., A materials perspective on Li-ion batteries at extreme temperatures, *Nat. Energy* 2 (8) (2017/07/24 2017) 17108, <https://doi.org/10.1038/nenergy.2017.108>.
- [9] W. Bai, X. Zhang, Z. Gao, et al., State of charge estimation for lithium-ion batteries under varying temperature conditions based on adaptive dual extended Kalman filter, *Elec. Power Syst. Res.* 213 (2022/12/01/2022) 108751, <https://doi.org/10.1016/j.epsr.2022.108751>.
- [10] J. Hu, Z. Yao, A. Chen, et al., High thermal-conductive phase change material by carbon fiber orientation for thermal management and energy conversion application, *Appl. Therm. Eng.* 265 (2025) 125566, <https://doi.org/10.1016/j.applthermaleng.2025.125566>, 2025/04/15/.
- [11] C.Y. Wang, G. Zhang, S. Ge, et al., Lithium-ion battery structure that self-heats at low temperatures, *Nature* 529 (7587) (Jan 28 2016) 515–518, <https://doi.org/10.1038/nature16502>.
- [12] S. Liu, G. Zhang, C.-Y. Wang, Challenges and innovations of lithium-ion battery thermal management under extreme conditions: a review, *ASME J. Heat Mass Trans.* 145 (8) (2023), <https://doi.org/10.1115/1.4056823>.
- [13] Z. Zou, J. Xie, Y. Luo, et al., Numerical study on a novel thermal management system coupling immersion cooling with cooling tubes for power battery modules, *J. Energy Storage* 83 (2024/04/01/2024) 110634, <https://doi.org/10.1016/j.est.2024.110634>.
- [14] A. Khan, S. Yaqub, M. Ali, et al., A state-of-the-art review on heating and cooling of lithium-ion batteries for electric vehicles, *J. Energy Storage* 76 (2024), <https://doi.org/10.1016/j.est.2023.109852>.
- [15] S.K. Mohammadian, Y. Zhang, Thermal management optimization of an air-cooled Li-ion battery module using pin-fin heat sinks for hybrid electric vehicles, *J. Power Sources* 273 (2015/01/01/2015) 431–439, <https://doi.org/10.1016/j.jpowsour.2014.09.110>.

- [16] D. Huang, Z. Chen, S. Zhou, Self-powered heating strategy for lithium-ion battery pack applied in extremely cold climates, *Energy* 239 (2022), <https://doi.org/10.1016/j.energy.2021.122095>.
- [17] R. Guo, L. Li, Z. Sun, et al., An integrated thermal management strategy for cabin and battery heating in range-extended electric vehicles under low-temperature conditions, *Appl. Therm. Eng.* 228 (2023/06/25/2023) 120502, <https://doi.org/10.1016/j.applthermaleng.2023.120502>.
- [18] Z.G. Qu, Z.Y. Jiang, Q. Wang, Experimental study on pulse self-heating of lithium-ion battery at low temperature, *Int. J. Heat Mass Tran.* 135 (2019) 696–705, <https://doi.org/10.1016/j.ijheatmasstransfer.2019.02.020>.
- [19] C. Lin, W. Kong, Y. Tian, et al., Sandwich self-heating structure-based lithium-ion battery system and its application in the fuel cell bus for Beijing Winter Olympic Games, *Energy Convers. Manag.* 284 (2023/05/15/2023) 116977, <https://doi.org/10.1016/j.enconman.2023.116977>.
- [20] J. Jiang, H. Ruan, B. Sun, et al., A low-temperature internal heating strategy without lifetime reduction for large-size automotive lithium-ion battery pack, *Appl. Energy* 230 (2018) 257–266, <https://doi.org/10.1016/j.apenergy.2018.08.070>.
- [21] H. Ruan, J. Jiang, B. Sun, et al., An optimal internal-heating strategy for lithium-ion batteries at low temperature considering both heating time and lifetime reduction, *Appl. Energy* 256 (2019), <https://doi.org/10.1016/j.apenergy.2019.113797>.
- [22] S. Guo, R. Xiong, K. Wang, et al., A novel echelon internal heating strategy of cold batteries for all-climate electric vehicles application, *Appl. Energy* 219 (2018) 256–263, <https://doi.org/10.1016/j.apenergy.2018.03.052>.
- [23] D.P. Finegan, J. Darst, W. Walker, et al., Modelling and experiments to identify high-risk failure scenarios for testing the safety of lithium-ion cells, *J. Power Sources* 417 (2019/03/31/2019) 29–41, <https://doi.org/10.1016/j.jpowsour.2019.01.077>.
- [24] H. Ruan, J. Jiang, B. Sun, et al., A rapid low-temperature internal heating strategy with optimal frequency based on constant polarization voltage for lithium-ion batteries, *Appl. Energy* 177 (2016/09/01/2016) 771–782, <https://doi.org/10.1016/j.apenergy.2016.05.151>.
- [25] M.B. Effat, C. Wu, F. Ciucci, Modeling efforts in the key areas of thermal management and safety of lithium ion battery cells: a mini review, *Asia Pac. J. Chem. Eng.* 11 (3) (2016/05/01 2016) 399–406, <https://doi.org/10.1002/apj.1999>.
- [26] Y. Qin, J. Du, L. Lu, et al., A rapid lithium-ion battery heating method based on bidirectional pulsed current: heating effect and impact on battery life, *Appl. Energy* 280 (2020/12/15/2020) 115957, <https://doi.org/10.1016/j.apenergy.2020.115957>.
- [27] R. Xiong, Z. Li, R. Yang, et al., Fast self-heating battery with anti-aging awareness for freezing climates application, *Appl. Energy* 324 (2022), <https://doi.org/10.1016/j.apenergy.2022.119762>.
- [28] H. Ruan, B. Sun, A. Cruden, et al., Optimal external heating resistance enabling rapid compound self-heating for lithium-ion batteries at low temperatures, *Appl. Therm. Eng.* 200 (2022), <https://doi.org/10.1016/j.applthermaleng.2021.117536>.
- [29] Y. Ji, C.Y. Wang, Heating strategies for Li-ion batteries operated from subzero temperatures, *Electrochim. Acta* 107 (2013/09/30/2013) 664–674, <https://doi.org/10.1016/j.electacta.2013.03.147>.
- [30] Z. Zhang, W. Yu, H. Li, et al., Heat transfer characteristics and low-temperature performance of a lithium-ion battery with an inner cooling/heating structure, *Appl. Therm. Eng.* 219 (2023/01/25/2023) 119352, <https://doi.org/10.1016/j.applthermaleng.2022.119352>.
- [31] H. Hailemichael, B. Ayalew, Combined film and pulse heating of lithium ion batteries to improve performance in low ambient temperature, *IFAC-PapersOnLine* 56 (3) (2023/01/01/2023) 427–432, <https://doi.org/10.1016/j.ifacol.2023.12.061>.
- [32] Q. Wang, B. Jiang, B. Li, et al., A critical review of thermal management models and solutions of lithium-ion batteries for the development of pure electric vehicles, *Renew. Sustain. Energy Rev.* 64 (2016/10/01/2016) 106–128, <https://doi.org/10.1016/j.rser.2016.05.033>.
- [33] A.P. Catalano, C. Scognamiglio, F. Piccirillo, et al., Thermo-electrochemical FEM and circuit simulations of Li-Ion batteries, *IEEE Trans. Compon. Packag. Manuf. Technol.* 13 (8) (2023) 1088–1095, <https://doi.org/10.1109/tcpmt.2023.3294702>.
- [34] W. Cao, Y. Qiu, P. Peng, et al., A full-scale electrical-thermal-fluidic coupling model for li-ion battery energy storage systems, *Appl. Therm. Eng.* 185 (2021), <https://doi.org/10.1016/j.applthermaleng.2020.116360>.
- [35] S. Panchal, M. Mathew, I. Dincer, et al., Thermal and electrical performance assessments of lithium-ion battery modules for an electric vehicle under actual drive cycles, *Elec. Power Syst. Res.* 163 (2018/10/01/2018) 18–27, <https://doi.org/10.1016/j.epr.2018.05.020>.
- [36] M.A. Perez Estevez, S. Calligaro, O. Bottesi, et al., An electro-thermal model and its electrical parameters estimation procedure in a lithium-ion battery cell, *Energy* 234 (2021), <https://doi.org/10.1016/j.energy.2021.121296>.
- [37] T. Mesbahi, R.B. Sugrañes, R. Bakri, et al., Coupled electro-thermal modeling of lithium-ion batteries for electric vehicle application, *J. Energy Storage* 35 (2021), <https://doi.org/10.1016/j.est.2021.102260>.
- [38] M. Gottapu, T. Goh, A. Kaushik, et al., Fully coupled simplified electrochemical and thermal model for series-parallel configured battery pack, *J. Energy Storage* 36 (2021), <https://doi.org/10.1016/j.est.2021.102424>.
- [39] K. Fu, T. Hamacher, V.S. Perić, Development of self-adaptive digital twin for battery monitoring and management system, *Elec. Power Syst. Res.* 234 (2024) 110698, <https://doi.org/10.1016/j.epr.2024.110698>.
- [40] C.X. He, Y.H. Liu, X.Y. Huang, et al., A reduced-order thermal runaway network model for predicting thermal propagation of lithium-ion batteries in large-scale power systems, *Appl. Energy* 373 (2024), <https://doi.org/10.1016/j.apenergy.2024.123955>.
- [41] A.P. Catalano, C. Scognamiglio, F. Piccirillo, et al., Analysis of the thermal behavior of Li-Ion pouch battery cell – part II: circuit-based modeling for fast and accurate thermo-electrochemical simulation. Presented at the 2022 28th International Workshop on Thermal Investigations of Ics and Systems (THERMINIC), 2022.
- [42] T. Huria, M. Ceraolo, J. Gazzarri, et al., High Fidelity electrical model with thermal dependence for characterization and simulation of high power lithium battery cells. Presented at the 2012 IEEE International Electric Vehicle Conference, 2012. Greenville, SC, USA.
- [43] J. E. Y. Qin, B. Zhang, et al., Effects of heating film and phase change material on preheating performance of the lithium-ion battery pack with large capacity under low temperature environment, *Energy* 284 (2023), <https://doi.org/10.1016/j.energy.2023.129280>.
- [44] C. Forgez, D. Vinh Do, G. Friedrich, et al., Thermal modeling of a cylindrical LiFePO₄/graphite lithium-ion battery, *J. Power Sources* 195 (9) (2010/05/01/2010) 2961–2968, <https://doi.org/10.1016/j.jpowsour.2009.10.105>.
- [45] K. Deep, K.P. Singh, M.L. Kansal, et al., A real coded genetic algorithm for solving integer and mixed integer optimization problems, *Appl. Math. Comput.* 212 (2) (2009) 505–518, <https://doi.org/10.1016/j.amc.2009.02.044>.
- [46] K. Saleem, K. Mehran, Z. Ali, Online reduced complexity parameter estimation technique for equivalent circuit model of lithium-ion battery, *Elec. Power Syst. Res.* 185 (2020/08/01/2020) 106356, <https://doi.org/10.1016/j.epr.2020.106356>.